Stylometric Analysis of Genre in Shakespearean Plays

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Goal

Could data analysis tools that are designed to solve matters of author attribution differentiate between Shakespeare's tragedies, comedies, and histories?

This would let us see whether or not there are consistent stylometric differences between Shakespeare plays of different genres.

Comedy	History	Tragedy
All's Well That Ends Well As You Like It The Comedy of Errors Cymbeline Love's Labours Lost Measure for Measure The Merry Wives of Windsor The Merchant of Venice A Midsummer Night's Dream Much Ado About Nothing Pericles, Prince of Tyre Taming of the Shrew The Tempest Troilus and Cressida Twelfth Night Two Gentlemen of Verona Winter's Tale	Henry IV, part 1 Henry IV, part 2 Henry V Henry VI, part 1 Henry VI, part 2 Henry VI, part 3 Henry VIII King John Richard II Richard III	Antony and Cleopatra Coriolanus Hamlet Julius Caesar King Lear Macbeth Othello Romeo and Juliet Timon of Athens Titus Andronicus

- 37 plays
  - 17 comedies
  - 10 histories
  - 10 tragedies

For testing I set aside 7 random plays

- 3 comedies
- 2 histories
- 2 tragedies

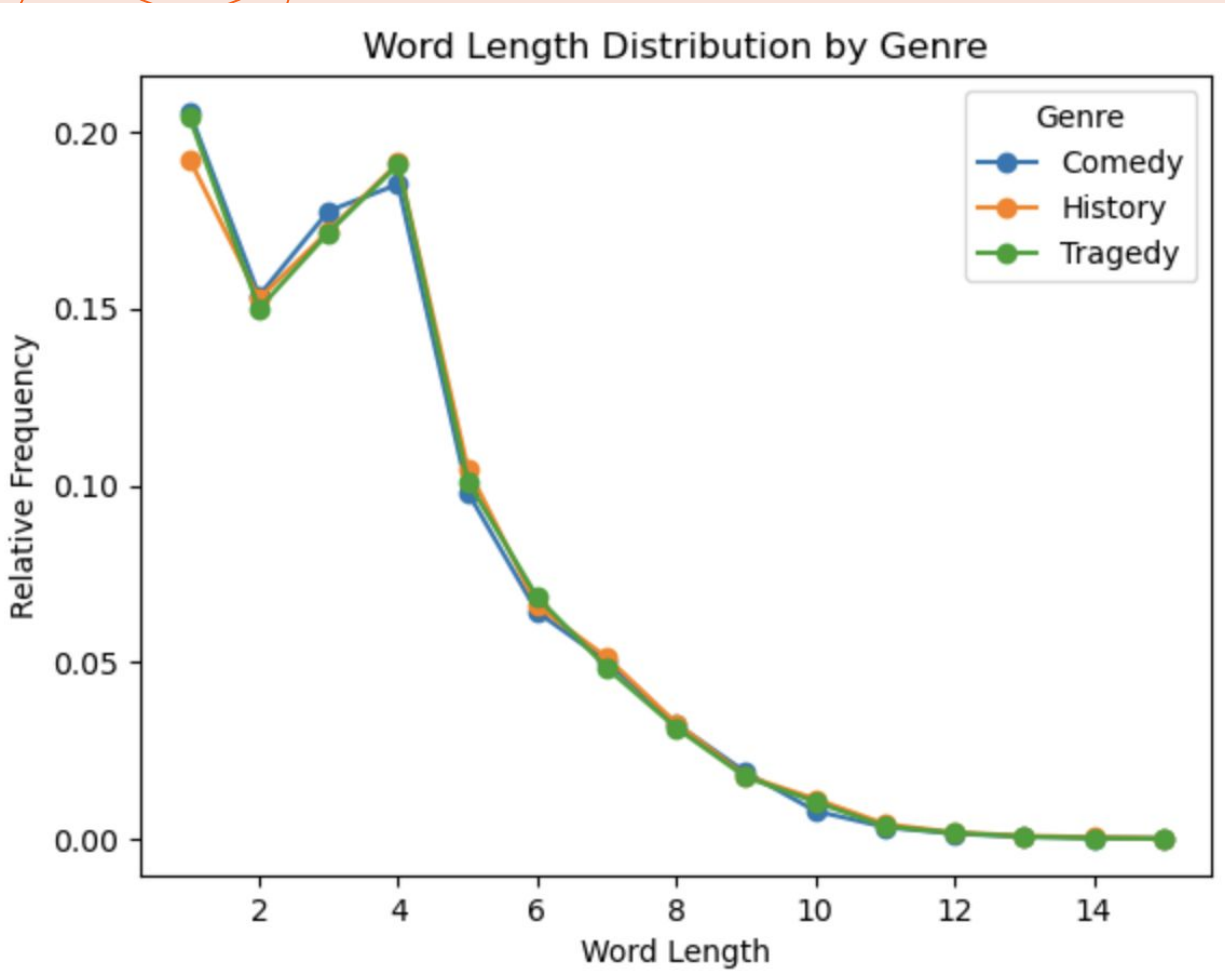
Method 1: Mendenhall’s Characteristic Curves of Composition

T. C. Mendenhall claimed that an author's stylistic signature could be found by measuring how often they used words of different lengths.

This metric is not very sophisticated but it is one of the first attempts at stylometry.

- Little variation
- Seems to hold no real predictive power

Mendenhall believed that an author's characteristic curve of composition taken from any large enough sample of their work would stay constant for their entire career.



Method 2: Kilgariff's Chi-Squared Method

Chi-squared is used to measure the distance between vocabulary used in different sets of text.

It is calculated by summing, over the n most common words, the squares of the differences between the actual numbers of tokens found in each author's corpus and the expected numbers, divided by the expected numbers.

Results

Test1
The Chi-squared statistic for Test1 and candidate Comedy is 5254.079718247389
The Chi-squared statistic for Test1 and candidate History is 7518.01678333913
The Chi-squared statistic for Test1 and candidate Tragedy is 6606.146304214536

Test2
The Chi-squared statistic for Test2 and candidate Comedy is 7893.470732592027
The Chi-squared statistic for Test2 and candidate History is 9456.54038407988
The Chi-squared statistic for Test2 and candidate Tragedy is 10602.34055104024

Test3
The Chi-squared statistic for Test3 and candidate Comedy is 4444.1364358131805
The Chi-squared statistic for Test3 and candidate History is 7390.803361682242
The Chi-squared statistic for Test3 and candidate Tragedy is 5959.956544678237

Test4
The Chi-squared statistic for Test4 and candidate Comedy is 4475.622663975111
The Chi-squared statistic for Test4 and candidate History is 3581.4660941600896
The Chi-squared statistic for Test4 and candidate Tragedy is 4530.849965425487

Test5
The Chi-squared statistic for Test5 and candidate Comedy is 5066.9544306573625
The Chi-squared statistic for Test5 and candidate History is 3928.872127308378
The Chi-squared statistic for Test5 and candidate Tragedy is 6917.883834115282

Th

This is the output for the first five tests

The smaller the chi-squared value the more similar the vocabulary is between the two corpora.

This test resulted in 7/7 accurate classifications.

Note about choosing n

"Scholars have suggested between 100 and 1,000 of the most common words; one project even used every word that appeared in the corpus at least twice. As a guideline, the larger the corpus, the larger the number of words that can be used as features without running the risk of giving undue importance to a word that occurs only a handful of times." (**) I tested 100, 200, 250, 300, 350, 500, 1000. 250 and 300 words gave me the highest accuracy.

100 words: 4/7
200 words: 6/7
250 words: 7/7
300 words: 7/7
350 words: 6/7
500 words: 4/7 *comedies misclassified
1000 words 4/7 *comedies misclassified

Method 3: John Burrow’s Delta Method

Burrows' Delta is designed to compare how a certain text and an arbitrary amount of sets of texts diverge from the average of them all put together.

It ends up taking the average of the absolute values of the differences between the z-scores for each feature between the anonymous paper and the candidate subcorpus.

The winning candidate is the genre for whom the delta score between the genre's subcorpus and the test case is the lowest.

The Delta Method gives equal weight to every feature that it measures, thus avoiding the problem of common words overwhelming the results, which was a potential issue with chi-squared tests.

Example of z-scores

Test case z-score for feature **the** is -0.2289715945413032
Test case z-score for feature **and** is 0.0791752161434511
Test case z-score for feature **i** is 1.5532975798697388
Test case z-score for feature **to** is -0.8972891161858926
Test case z-score for feature **of** is -0.0990939719915438 1

Z-scores for Test1 and 5 most common words in collective corpus

Example of Delta score output

Test1
Delta score for Test1 and candidate History is 1.269229452512254
Delta score for Test1 and candidate Tragedy is 1.098408253463023
Delta score for Test1 and candidate Comedy is 0.7573992753080779
Delta score for Test1 and candidate Test1 is 0.0
Delta score for Test1 and candidate Test2 is 1.0725140041237828
Delta score for Test1 and candidate Test3 is 1.0330376340838199
Delta score for Test1 and candidate Test4 is 1.1465872885963129
Delta score for Test1 and candidate Test5 is 1.8186567864693626
Delta score for Test1 and candidate Test6 is 0.8834041835187684
Delta score for Test1 and candidate Test7 is 1.4387657747912805

We can see the similarity between Test1 and each genre as well as the difference between Test1 and the other tests.

Results

Correct: 6/7

Test6, which was Julius Caesar, has a delta score of .8010 for comedy and a delta score of .8124 for tragedy. Under the delta method it is classified as comedy but it is assigned a similar difference from both comedy and tragedy.

Further analysis

I calculated the z-scores for each genre and the 20 most frequent words. I then looked to see which z-scores had an absolute value greater than 0.75. This demonstrated some words each genre used more ‘abnormally’.

Comedy

and is -1.2222777747298392
i is 0.8624952348036443
with is -0.9911647217116532

History

you is -0.9315821618474616
is is -1.678440195325424
not is -1.3167595685477231
it is -1.1825624171672608
his is 0.9839694634869856

Tragedy

be is -0.8215327168620244

Conclusion

Ultimately, I was able to use methods designed to identify authorship to prove that there is a stylistic difference between Shakespeare's comedies, histories, and tragedies.

The genres seemed not to have varied very distinctly when observing Mendenhall's characteristic curves of composition, which displays the frequency of words of certain lengths. However, while Shakespeare kept the length of words somewhat constant, the genres do seem distinguishable when looking at vocabulary.

More analysis could be done on what features of Shakespeare’s vocabulary enable this distinguishability.